DL-SIFA: Deep Learning Statistical Ineffective Fault Analysis on AES-128

CS6630 — Secure Processor Microarchitecture

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Outline

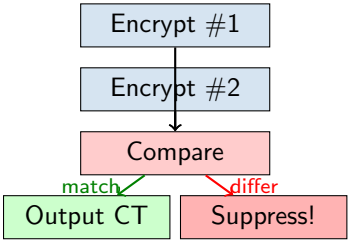
Background: Fault Attacks and Defenses

Fault Attacks:

- Inject physical faults during encryption
- Faulty output leaks the secret key
- DFA needs both correct and faulty ciphertexts

Defense — Temporal Redundancy:

- Encrypt the same plaintext **twice**, compare
- Outputs differ $\rightarrow$ fault detected $\rightarrow$ suppress
- Blocks DFA completely



SIFA: Breaking Temporal Redundancy

Key Insight (Dobraunig et al., TCHES 2018)

$\sim$50% of injected faults are *ineffective*—the target bit was already in the faulted state. These pass the redundancy check, but carry a **statistical bias** that leaks the key.

Project Goal:

1. Implement AES-128 with temporal redundancy + stuck-at-0 faults at Round 8
2. **Traditional SIFA**: recover key using bit-bias formula
3. **DL-SIFA**: recover key using a trained neural network
4. Compare trace efficiency of both methods

Based on: Ramezanpour et al., *"Fault Intensity Map Analysis with NN Key Distinguisher"*, ASHES 2019.

Fault Model and Bias Mechanism

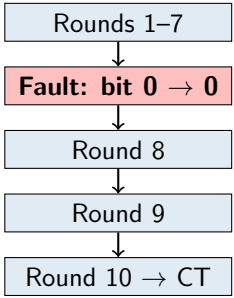
Fault	Stuck-at-0 (bit 0)
Location	Round 8 SubBytes input
Attempts	50,000 per byte
Ineffective rate	$\sim$50%
Traces kept	$\sim$25,000

Why bias?

Normal: byte $\in \{0 \ldots 255\}$ (256 values)

After fault: only even values survive (128)

$\rightarrow$ half the histogram bins are empty



Traditional SIFA: Bit-Bias Key Recovery

Algorithm: For each key guess $k \in \{0, \ldots, 255\}$:

1. Invert each CT through rounds 10, 9, 8 using guess k
2. Compute bias $= \frac{\#(\text{bit } 0{=}0)}{N}$. Correct key $\rightarrow 1.0$, Wrong $\rightarrow 0.5$

Results:

- All 16 bytes: bias $= 1.0000$
- Recovered $K_{10}[0] = 0\text{xD0}$ ✓
- Correct bias: 1.0, avg wrong: 0.498
- **Min traces: 20**

Progressive recovery:

Traces	Result
10	FAIL
20	**SUCCESS**
50+	SUCCESS

DL-SIFA: Neural Network Approach

Motivation: Traditional SIFA needs exact fault model. In real hardware, faults are noisy and imprecise.

Profiling (500 random keys):

1. Collect faulted + clean CTs
2. Invert to R8, build **256-bin histogram**
3. Label: biased = 1, clean = 0
4. 3,000 training samples

Attack (unknown key):

1. For each guess: invert, histogram, NN score
2. Highest score = recovered key

MLP Architecture:

Layer	Size	Details
Input	256	Histogram
FC+BN	128	ReLU, Drop 0.3
FC+BN	64	ReLU, Drop 0.2
FC	32	ReLU
Output	1	Sigmoid

BCE loss, Adam (lr=0.001)
200 epochs, PyTorch

DL-SIFA: Results

Training: 100% accuracy

Key recovery:

- Recovered $K_{10}[0] = \texttt{0xD0}$ ✓
- Correct score: **1.0000**
- Avg wrong score: 0.0001
- **Min traces: 50**

Stronger separation than traditional:
1.0 vs 0.0 (instead of 1.0 vs 0.5)

Progressive recovery:

Traces	Result
10	FAIL
20	FAIL
50	**SUCCESS**
100+	SUCCESS

With $<$50 traces, histogram is too noisy for the NN.

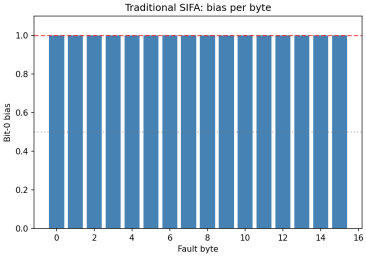
Comparison: Traditional vs DL-SIFA

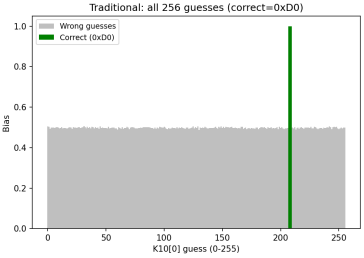
Criterion	Traditional SIFA	DL-SIFA
Key recovered	✓(0xD0)	✓(0xD0)
Min traces	**20**	50
Separation	1.0 vs 0.5	**1.0 vs 0.0**
Needs fault model	Yes	**No**
Noise tolerance	No	**Yes**
Complexity	Low (formula)	High (training)

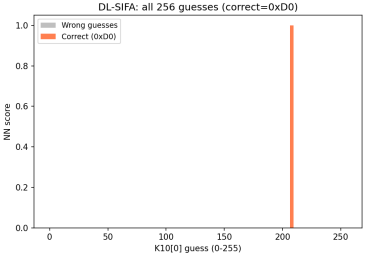
Traditional: fewer traces, simpler, but only works with known fault model

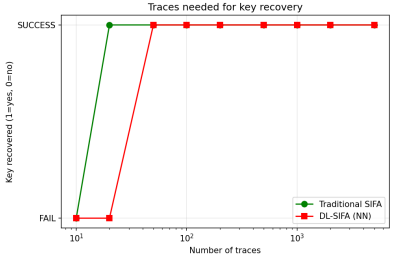
DL-SIFA: stronger signal, works without knowing fault details, more practical for real hardware

Results


Traditional SIFA: bias per byte
Bit-0 bias
Fault byte


Traditional: all 256 guesses (correct=0xD0)
Wrong guesses
Correct (0xD0)
Bias
K10[0] guess (0-255)


DL-SIFA: all 256 guesses (correct=0xD0)
Wrong guesses
Correct (0xD0)
NN score
K10[0] guess (0-255)


Traces needed for key recovery
SUCCESS
FAIL
Key recovered (1=yes, 0=no)
Traditional SIFA
DL-SIFA (NN)
Number of traces

Conclusion

1. SIFA breaks AES-128 with temporal redundancy using only **correct outputs**
2. Both methods recover $K_{10}[0] = \texttt{0xD0}$
3. Traditional SIFA: **20 traces**, but needs exact fault model
4. DL-SIFA: **50 traces**, fault-model agnostic, better separation
5. Real-world faults are rarely known precisely $\rightarrow$ DL-SIFA is more practical

Future work: noisy faults, multi-bit faults, other ciphers, CNNs

References:

1. Dobraunig et al., "SIFA," TCHES 2018
2. Ramezanpour et al., "Fault intensity map analysis with NN key distinguisher," ASHES 2019

Thank You!

Questions?